

CHAPTER I

INTRODUCTION

1.1 Background and Motivation

Air quality degradation is one of the most pressing public health concerns of the contemporary era. According to the World Health Organization (WHO, 2021), approximately 99% of the global population breathes air that exceeds WHO guideline limits, with low- and middle-income countries bearing a disproportionately high burden. In India, the problem is particularly acute: the country accounts for a significant share of global premature deaths attributable to outdoor air pollution, with major metropolitan areas consistently registering Air Quality Index (AQI) values classified as "Poor," "Very Poor," or "Severe" for extended periods.

Delhi, the National Capital Territory of India, is widely regarded as one of the most polluted capital cities in the world. Its unique combination of geographic, demographic, and industrial characteristics creates a perfect storm of air quality deterioration. The city is home to over 20 million residents, hosts one of the densest vehicular fleets in Asia, and is located in the Indo-Gangetic Plain — a region characterised by topographic and meteorological conditions that impede the dispersal of pollutants. During winter months (October–February), the combination of crop stubble burning from neighbouring states of Punjab and Haryana, reduced wind speeds, atmospheric temperature inversions, and concentrated vehicular and industrial emissions drives AQI values to critically hazardous levels.

The quantification and prediction of air quality is therefore a matter of urgent scientific and policy relevance. Traditional air quality monitoring systems in India rely on a network of Continuous Ambient Air Quality Monitoring Stations (CAAQMS) operated by the Central Pollution Control Board (CPCB) and State Pollution Control Boards. While these stations provide real-time data, the raw concentration measurements are most useful to public health agencies and the general population when translated into the standardised AQI framework, which maps complex multi-pollutant information into six intuitive categories (from "Good" to "Severe") accompanied by colour codes and health advisories.

The application of machine learning (ML) to air quality prediction represents a natural evolution of this monitoring infrastructure. ML classifiers can learn the nonlinear, highdimensional relationships between pollutant concentrations and AQI categories from

historical data, enabling automated, real-time AQI category assignment that is both faster and potentially more robust than rule-based computation alone. Furthermore, ML models can identify the most important predictors of air quality deterioration, providing actionable insights for policy design.

1.2 The Air Quality Index Framework

The Air Quality Index (AQI) is a standardised metric developed by regulatory agencies to communicate air quality conditions to the public in a simple, interpretable format. India's National AQI was established by the CPCB in 2014 (CPCB, 2014) and is based on the subindex approach: for each of eight pollutants (PM_{2.5}, PM₁₀, NO₂, SO₂, CO, O₃, NH₃, and Pb), a sub-index is computed by linear interpolation within predefined concentration breakpoints. The overall AQI is taken as the maximum sub-index value across all measured pollutants.

The six AQI categories and their associated health implications are presented in Table 1.1 below. The thresholds reflect the CPCB's assessment of concentration levels at which adverse health effects begin to manifest across the general population, with stricter thresholds for sensitive groups including children, the elderly, and individuals with pre-existing respiratory or cardiovascular conditions.

Table 1.1: CPCB AQI Categories with PM_{2.5} Breakpoints and Health Implications

AQI Category	AQI Range	PM _{2.5} (µg/m ³)	Health Implication
Good	0–50	0–30	Minimal impact
Satisfactory	51–100	31–60	Minor discomfort to sensitive people
Moderate	101–200	61–120	Discomfort to people with lung/heart disease
Poor	201–300	121–250	Discomfort to most people, health alert
Very Poor	301–400	251–380	Respiratory illness with prolonged exposure
Severe	401–500	>380	Health impact even with light exposure

1.3 Problem Statement

Despite the availability of real-time concentration data from CAAQMS networks, several challenges remain in the operational deployment of AQI classification:

- Manual AQI computation is rule-based and does not incorporate learning from historical patterns or contextual factors such as season and temporal trends.
- The nonlinear interactions between pollutants — particularly PM_{2.5}, PM₁₀, NO₂, and CO — are not fully captured by the maximum sub-index rule, which may lead to misclassification in pollution episodes characterised by multiple simultaneously elevated pollutants.
- There is a lack of systematic quantitative analysis of how well different machine learning architectures perform on the AQI classification task using large-scale, multistation Delhi data.
- The feature importance of different pollutants in determining AQI category — from a data-driven perspective — has not been comprehensively characterised for the 2020–2024 period.

This study addresses these gaps by developing and rigorously evaluating a suite of ML classifiers on a large, diverse, multi-station dataset, providing a data-driven characterisation of Delhi's air quality dynamics over five years.

1.4 Objectives of the Study

The principal objectives of this study are:

1. To construct and rigorously clean a multi-station, multi-pollutant air quality dataset for Delhi covering the period January 2020 to December 2024, comprising 60,291 station-day observations.
2. To perform comprehensive exploratory data analysis (EDA) including descriptive statistics, distributional analysis, temporal trend analysis, seasonal decomposition, and spatial analysis of AQI patterns across monitoring stations.
3. To develop and implement five supervised machine learning classifiers — Random Forest, Gradient Boosting, Logistic Regression, Decision Tree, and K-Nearest Neighbours — for the multi-class AQI category prediction task.
4. To evaluate model performance using Accuracy, Weighted F1-Score, Precision, Recall, and class-wise F1-Scores, and to identify the best-performing model for operational deployment.
5. To quantify the relative importance of different pollutants in determining AQI category using Random Forest feature importance analysis.

6. To provide policy-relevant conclusions and deployment recommendations for realtime AQI alert systems.

1.5 Scope and Delimitations

This study focuses on the AQI classification problem at the station-day level, using concurrent pollutant concentration measurements as predictors. The scope is deliberately delimited to the classification task rather than the forecasting task (predicting future AQI from past observations), which is substantially more challenging and requires additional meteorological inputs. The study covers 25 CPCB monitoring stations across Delhi for the fiveyear period 2020–2024, providing both temporal depth and spatial breadth.

Meteorological variables (temperature, humidity, wind speed, boundary layer height) are not included as predictors in the present study, as the focus is on characterising the pollutant-to-AQI relationship. Future work should explore the incremental contribution of meteorological predictors to classification accuracy.

1.6 Significance of the Study

The significance of this research is threefold. First, it provides the most comprehensive ML-based analysis of Delhi's AQI using five years of post-2020 data, a period that includes the COVID-19 lockdown as a natural experiment in pollution reduction. Second, it delivers actionable deployment recommendations, including a ranked comparison of five classifiers and a learning curve analysis that guides data collection strategy. Third, the feature importance analysis provides quantitative evidence for pollutant-specific policy priorities, reinforcing the case for PM_{2.5}-targeted emission control measures.

1.7 Organisation of the Report

The remainder of this report is organised as follows. Chapter II reviews the relevant literature on air quality prediction using statistical and machine learning methods. Chapter III describes the dataset, preprocessing pipeline, and methodological framework in detail. Chapter IV presents the results of the EDA and model evaluation. Chapter V summarises the key findings, draws policy-relevant conclusions, and outlines directions for future research.

CHAPTER II

REVIEW OF LITERATURE

2.1 Statistical Approaches to Air Quality Prediction

The earliest quantitative approaches to air quality prediction were rooted in classical time series methods. AutoRegressive Integrated Moving Average (ARIMA) and its seasonal extension (SARIMA) models were widely applied to pollutant concentration series in the 1990s and 2000s Siew et al., (2008); Chaudhuri & Dutta, (2014). These models are well-suited to capturing autocorrelation and seasonality in stationary time series, and their interpretability is a significant advantage in policy contexts. However, their linear formulation limits their ability to model the complex, nonlinear interactions between multiple pollutants that drive AQI dynamics.

Box-Jenkins ARIMA models were applied to AQI series in Malaysian cities by Siew et al. (2008), achieving reasonable short-term forecasting accuracy but struggling with extreme events during haze episodes. Chaudhuri and Dutta (2014) applied ARIMA models to pollutant series in Kolkata, India, and demonstrated that seasonal patterns are the dominant driver of predictability, but that residual nonlinearity remains after ARIMA fitting. Gopu et al. (2021) applied ARIMA to Hyderabad air quality data, confirming the utility of time series methods for Indian urban contexts.

Regression-based approaches have also been widely employed. Slini et al. (2002) used multiple regression to model AQI in Greek cities, identifying wind speed and traffic density as key predictors. Pakrooh and Pishbahar (2019) developed hybrid regression models for Iranian cities. However, the limitations of linearity and the assumption of Gaussian residuals motivate the transition to machine learning methods for complex, high-dimensional air quality datasets.

2.2 Machine Learning Methods for AQI Classification

The application of machine learning to air quality classification accelerated significantly in the 2010s, driven by the increasing availability of large-scale monitoring datasets and computational resources. Random Forest Breiman, (2001) and Gradient Boosting Friedman, (2001) have emerged as the dominant ensemble methods for AQI-related classification and regression tasks.

Random Forest builds a large ensemble of decision trees on bootstrapped subsets of the training data, with feature randomisation at each split. This combination of bagging and feature randomisation produces a model that is highly resistant to overfitting and provides reliable feature importance estimates through mean decrease in impurity. Gradient Boosting constructs trees sequentially, with each tree fitted to the residuals of the previous ensemble, producing a powerful nonlinear model that often matches or exceeds Random Forest performance on structured tabular data.

Several studies have demonstrated the superiority of ensemble methods over single classifiers for AQI prediction. Liu and You (2022) applied both ARIMA and Random Forest to Beijing AQI data, demonstrating that Random Forest substantially outperforms ARIMA for the classification task while ARIMA remains competitive for short-term forecasting. Ye (2019) compared multiple ML models for Shenzhen air quality prediction, confirming the strong performance of tree-based ensembles.

Logistic Regression, despite its simplicity, has been shown to achieve competitive performance on the AQI classification task when the decision boundaries between categories are approximately linear in the pollutant concentration space. Its interpretability — through examination of coefficient weights — and its ability to produce calibrated probability estimates make it a useful baseline and production model in resource-constrained deployments.

K-Nearest Neighbours (KNN) is a non-parametric, instance-based classifier that has been applied to air quality classification with mixed results. Its sensitivity to the curse of dimensionality and its high memory and inference cost at large scale are significant drawbacks, but it provides a useful non-parametric baseline that captures local structure in the feature space.

2.3 Deep Learning Approaches

Deep learning architectures, particularly Long Short-Term Memory (LSTM) networks and Transformer-based models, have demonstrated strong performance on the air quality forecasting task Arsov et al., (2020). LSTMs are well-suited to multi-step-ahead pollutant forecasting due to their ability to capture long-range temporal dependencies. However, for the concurrent classification task (predicting the current AQI category from current pollutant measurements), the advantage of deep learning over well-tuned ensemble methods is less clear, and the interpretability trade-off is more significant.

The hybrid approach of combining ARIMA with neural networks, pioneered by Samia et al. (2012) for Tunisian cities, has been applied to Indian urban contexts with promising results. These hybrid models can capture both linear temporal structure (ARIMA component) and nonlinear spatial-pollutant interactions (neural network component).

2.4 Studies Specific to Delhi and the Indo-Gangetic Plain

Delhi's air quality has attracted significant research attention due to its severity and public health consequences. Several studies have documented the strong seasonal pattern of AQI in Delhi, with winter months consistently exhibiting the highest AQI values and the highest frequency of Severe episodes. The convergence of stubble burning emissions, reduced planetary boundary layer height, and temperature inversions during October–February creates a synergistic pollution amplification that is particularly difficult to predict from pollutant concentrations alone. The COVID-19 lockdown of March–May 2020 provided an unprecedented natural experiment: with vehicular and industrial activity reduced to minimal levels, Delhi's AQI dropped sharply, providing a quantitative estimate of the contribution of these emission sources to ambient air quality. This natural experiment is captured in the present dataset and is reflected in the lower yearly average AQI observed in 2020 compared to subsequent years. The spatial heterogeneity of Delhi's air quality has also been documented. Industrial areas (particularly in the north and west of the city) and traffic-dense corridors consistently exhibit higher AQI values than green belt areas and southern residential zones. This spatial variation motivates the station-level analysis presented in Chapter IV.

2.5 Research Gap and Contribution

The reviewed literature reveals several gaps that the present study addresses. First, most existing ML-based AQI studies focus on individual stations or city-aggregated data; the present study uses a multi-station dataset covering 25 monitoring points, providing a more comprehensive spatial picture. Second, few studies have rigorously compared five ML classifiers on the full six-class CPCB AQI classification problem using post-2020 data. Third, the learning curve analysis presented in Chapter IV, which characterises the data efficiency of the best model, has not been previously reported for this dataset. Finally, the present study provides a comprehensive spatial analysis of station-level AQI patterns and PM_{2.5}/PM₁₀ ratios by station type, contributing to the characterisation of Delhi's pollution source mix.

CHAPTER III

METHODOLOGY

3.1 Dataset Description

The dataset used in this study is the Delhi Air Quality Dataset (2020–2024, All Stations), comprising daily air quality measurements from 33 Continuous Ambient Air Quality Monitoring Stations (CAAQMS) operated by the Central Pollution Control Board (CPCB) across the National Capital Territory of Delhi. The dataset contains 60,291 station-day observations spanning January 1, 2020 to December 31, 2024, covering five complete calendar years.

3.1.1 Data Collection and Source

Data were sourced from the CPCB's real-time air quality monitoring portal (<https://app.cpcbcr.com/>). Each record corresponds to a single station-day observation and includes the following variables: date, year, month, day, season (categorical), station name, zone (administrative area of Delhi), geographic coordinates (latitude, longitude), concentrations of six pollutants (PM_{2.5}, PM₁₀, NO₂, SO₂, CO, O₃), the computed AQI value, and the AQI category label as defined by the CPCB.

3.1.2 Monitoring Station Network

The 25 monitoring stations span all major administrative zones of Delhi, including East Delhi (Anand Vihar, Patparganj), North Delhi (Rohini, Wazirpur, Jahangirpuri), South Delhi (R.K. Puram, Lodhi Road), West Delhi (Punjabi Bagh, Dwarka), Central Delhi (ITO, Mandir Marg), and additional stations in industrial, traffic, and green belt locations. The geographic coordinates of each station are recorded in the dataset, enabling the spatial analysis presented in Section 4.6.

Stations are further classified by type: Airport, Green (park/forest), Industrial, Traffic, and Other (residential/institutional). This classification is used in the PM_{2.5}/PM₁₀ ratio analysis to characterise the dominant emission sources by station type.

3.1.3 Feature Variables

The six pollutant concentration variables and the calendar month constitute the predictor features used in all ML models. Their units and roles are described below:

Table 3.1: Feature Variables Used in Machine Learning Models

Variable	Unit	Description	Role
PM2.5	$\mu\text{g}/\text{m}^3$	Fine particulate matter ($\leq 2.5 \mu\text{m}$ diameter)	Primary predictor
PM10	$\mu\text{g}/\text{m}^3$	Coarse particulate matter ($\leq 10 \mu\text{m}$ diameter)	Primary predictor
NO ₂	$\mu\text{g}/\text{m}^3$	Nitrogen dioxide	Primary predictor
SO ₂	$\mu\text{g}/\text{m}^3$	Sulfur dioxide	Primary predictor
CO	mg/m^3	Carbon monoxide	Primary predictor
O ₃	$\mu\text{g}/\text{m}^3$	Ground-level ozone	Primary predictor
Month	1–12 (integer)	Calendar month of observation	Temporal predictor
AQI_Category	Categorical	CPCB AQI category label (6 classes)	Target variable

3.2 Data Preprocessing

The raw dataset was subjected to a rigorous preprocessing pipeline to ensure data quality, consistency, and suitability for machine learning. The following steps were applied sequentially:

3.2.1 Missing Value Treatment

A systematic audit of missing values was conducted for each pollutant variable across all 25 stations. The overall missing value rate was below 3% for PM2.5, PM10, and NO₂, and below 5% for SO₂, CO, and O₃. Missing values were imputed using the station-specific monthly median imputation strategy: for each missing observation at a given station and month, the median value of that pollutant across all non-missing observations from the same station and same month was used as the imputed value. This approach preserves both the station-specific baseline and the seasonal pattern of each pollutant.

Records with more than three pollutant values missing simultaneously were excluded from the analysis, as imputation of multiple simultaneous missing values would introduce excessive uncertainty. A total of 47 records (0.08% of the dataset) were removed on this basis.

3.2.2 Outlier Detection and Treatment

Outliers were identified using the Interquartile Range (IQR) method. For each pollutant p , the upper fence was defined as:

$$\text{Upper Fence} = Q_3(p) + 3.0 \times IQR(p)$$

where $Q_3(p)$ is the 75th percentile and $IQR(p) = Q_3(p) - Q_1(p)$ is the interquartile range of pollutant p . A multiplier of 3.0 (rather than the standard 1.5) was used to preserve genuine extreme pollution episodes (e.g., Diwali-related PM_{2.5} spikes) while identifying physically implausible measurement errors.

Records exceeding the upper fence for PM_{2.5} ($> 515 \mu\text{g}/\text{m}^3$) or PM₁₀ ($> 963 \mu\text{g}/\text{m}^3$) were reviewed individually. Those coinciding with known extreme pollution events (e.g., Diwali, stubble burning peaks) were retained. Those not associated with any known event were replaced with the station-specific 99th percentile value. This winsorisation approach preserves the distributional shape while capping physically implausible values.

3.2.3 Data Transformation

Beyond the six raw pollutant concentrations, the calendar month (integer 1–12) was included as an additional feature to capture the strong seasonal pattern in Delhi's air quality. No polynomial or interaction features were engineered, as the tree-based models used in the study (Random Forest and Gradient Boosting) are capable of learning nonlinear interactions directly from the raw features. The AQI category label was encoded as an ordinal integer for KNN and Logistic Regression training (Good=0, Satisfactory=1, Moderate=2, Poor=3, Very Poor=4, Severe=5), while tree-based models used the string labels directly.

3.2.4 Train-Test Split

The dataset was split into training (80%, $n = 48,232$) and test (20%, $n = 12,059$) sets using stratified random sampling to ensure that the proportion of each AQI category in the training and test sets matches the overall distribution. Stratification is critical given the class imbalance in the dataset, particularly for the rare "Good" (approximately 1%) and "Severe"

(approximately 2%) categories. A fixed random seed (seed = 42) was used for all random operations to ensure full reproducibility of results.

3.3 Machine Learning Models

3.3.1 Decision Tree Classifier

A Decision Tree classifier recursively partitions the feature space into rectangular regions by selecting splits that maximise the reduction in node impurity. The Gini Impurity at a node t is defined as:

$$Gini(t) = 1 - \sum (p(c|t))^2$$

where $p(c|t)$ is the proportion of training observations at node t belonging to class c , and the sum is over all C classes. At each split, the feature j and threshold θ are selected to maximise the weighted impurity reduction:

$$\Delta Gini(t, j, \theta) = Gini(t) - \left(\frac{N_l}{N}\right) \times Gini(t_l) - \left(\frac{N_r}{N}\right) \times Gini(t_r)$$

where N is the number of observations at node t , N_l and N_r are the numbers of observations in the left and right child nodes t_l and t_r , respectively. Trees are grown to maximum depth unless regularised by minimum samples per leaf and maximum depth hyperparameters.

3.3.2 Random Forest

Random Forest (Breiman, 2001) constructs B decision trees on B bootstrapped resamples of the training data. At each node in each tree, only a random subset of $\sqrt{m}$ features (where m is the total number of features) is considered for the split. The final prediction is determined by majority vote across all B trees:

$$\hat{y}(x) = \arg \max_c \sum_{b=1}^B \mathbb{1}(h^b(x) = c)$$

where $\mathbb{1}^b(x)$ is the indicator function that tree b predicts class c for input x . The use of bootstrapping (bagging) and feature randomisation reduces the correlation between trees and therefore reduces the variance of the ensemble prediction relative to a single tree. Feature importance in Random Forest is measured by the Mean Decrease in Impurity (MDI):

$$Importance(j) = \left(\frac{1}{B}\right) \sum_{b=1}^B \sum_t \Delta Gini(t, j, \theta^*) \times \left(\frac{N_t}{N}\right)$$

Where the inner sum is over all nodes t in tree b that split on feature j , and $\Delta Gini$ is the impurity reduction at that node.

3.3.3 Gradient Boosting

Gradient Boosting (Friedman, 2001) constructs an additive ensemble of trees by sequentially fitting each new tree to the negative gradient of the loss function with respect to the current ensemble prediction. For the multi-class cross-entropy loss, the ensemble at iteration m is:

$$Fm(x) = Fm-1(x) + \eta \times hm(x)$$

where $Fm(x)$ is the ensemble prediction after m trees, η is the learning rate (shrinkage parameter), and $hm(x)$ is the m th tree fitted to the pseudo-residuals. The learning rate η controls the trade-off between the number of trees B and generalisation: smaller η requires more trees but typically yields better generalisation. In this study, $\eta = 0.1$ and $B = 100$ were used.

3.3.4 Multinomial Logistic Regression

For the six-class AQI classification problem, Logistic Regression uses the Softmax function to model the conditional probability of each class:

$$P(y = c | x) = \frac{\exp(w_c^T x + b_c)}{\sum_{k=1}^6 \exp(w_k^T x + b_k)}$$

where w_r is the weight vector for class c , b_r is the bias term, and the sum in the denominator is over all $K = 6$ classes.

3.3.5 K-Nearest Neighbours (KNN)

K-Nearest Neighbours is a non-parametric instance-based classifier. For a test observation x , the predicted class is determined by majority vote among the K nearest training observations in the feature space:

$$\hat{y}(x) = \arg \max_c \sum_{i \in N_K(x)} \mathbb{1}(y_i = c)$$

where $N_K(x)$ is the set of K nearest neighbours of x , and $\mathbb{1}(\cdot)$ is the indicator function. Euclidean distance was used as the distance metric, with $K = 5$ selected by cross-validation.

All features were standardised (zero mean, unit variance) prior to KNN fitting to ensure that pollutants measured in different units (e.g., PM2.5 in $\mu\text{g}/\text{m}^3$ vs. CO in mg/m^3) do not dominate the distance computation.

3.4 Model Evaluation Metrics

Model performance was assessed using four complementary metrics computed on the held-out test set:

3.4.1 Accuracy

Accuracy is the fraction of test observations correctly classified across all six AQI categories:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Predictions}} = \frac{TP_{\text{total}}}{N_{\text{test}}}$$

While intuitive, accuracy can be misleading in the presence of class imbalance (e.g., if a model simply predicts "Moderate" for all observations, it would achieve approximately 49.7% accuracy). It is therefore reported alongside F1-scores.

3.4.2 Precision, Recall, and F1-Score

For each class c , precision, recall, and F1-score are defined as:

$$\text{Precision}(c) = \frac{TP_c}{TP_c + FP_c}$$

$$\text{Recall}(c) = \frac{TP_c}{TP_c + FN_c}$$

$$F1(c) = 2 \times \frac{\text{Precision}(c) \times \text{Recall}(c)}{(\text{Precision}(c) + \text{Recall}(c))}$$

where TP_r , FP_r , and FN_r are the true positives, false positives, and false negatives for class c , respectively.

CHAPTER IV

RESULTS AND INTERPRETATION

4.1 Descriptive Statistics

Table 4.1 presents the descriptive statistics for all six pollutant concentration variables and the AQI, computed over all 60,291 station-day observations in the cleaned dataset. The statistics reveal several important characteristics of Delhi's pollution profile.

Table 4.1: Descriptive Statistics of Key Pollutants (2020–2024)

Pollutant	Count	Mean	Std Dev	Min	Q1 (25%)	Median (50%)	Q3 (75%)	Max
PM2.5 ($\mu\text{g}/\text{m}^3$)	60,291	101.14	64.12	5.0	53.2	84.3	137.0	515.2
PM10 ($\mu\text{g}/\text{m}^3$)	60,291	166.92	107.44	10.0	86.8	138.5	225.2	963.6
NO ₂ ($\mu\text{g}/\text{m}^3$)	60,291	65.58	40.36	5.0	35.1	55.1	88.8	323.7
SO ₂ ($\mu\text{g}/\text{m}^3$)	60,291	21.44	14.09	2.0	11.0	17.8	29.1	127.2
CO (mg/m^3)	60,291	2.15	1.34	0.2	1.15	1.79	2.90	11.19
O ₃ ($\mu\text{g}/\text{m}^3$)	60,291	40.02	15.16	5.0	29.3	39.2	50.0	128.8
AQI	60,291	170.41	57.13	21.0	145.0	166.0	193.0	500.0

The mean PM2.5 concentration of 101.14 $\mu\text{g}/\text{m}^3$ is approximately 3.4 times the WHO 24-hour guideline of 15 $\mu\text{g}/\text{m}^3$ and substantially exceeds even the CPCB's own 24-hour standard of 60 $\mu\text{g}/\text{m}^3$. The high standard deviation (64.12 $\mu\text{g}/\text{m}^3$) reflects the enormous variability between clean monsoon days and severely polluted winter episodes. PM10 shows similar characteristics, with a mean of 166.92 $\mu\text{g}/\text{m}^3$ and a maximum of 963.6 $\mu\text{g}/\text{m}^3$ — nearly ten times the WHO guideline of 45 $\mu\text{g}/\text{m}^3$. The mean AQI of 170.41 falls in the "Moderate" category (101–200), but the median of 166 and the 75th percentile of 193 indicate that approximately half of all station-days have AQI values close to or at the upper boundary of Moderate, with a substantial tail extending into the Poor and Very Poor categories.

4.2 AQI Value Distribution

Figure 4.1 shows the distribution of AQI values across all 60,291 observations, colourcoded by AQI category. The distribution is right-skewed with a dominant peak in the Moderate range (AQI 101–200), reflecting the predominance of this category across all seasons and stations. The distribution has a pronounced shoulder in the Poor range (AQI 201–300) and a long right tail extending to Severe (AQI > 400) episodes that, while rare in frequency, represent the most health-critical pollution events. The Good category (AQI < 50) appears as a small peak at the left tail of the distribution, confirming that clean air days are extremely rare in Delhi.

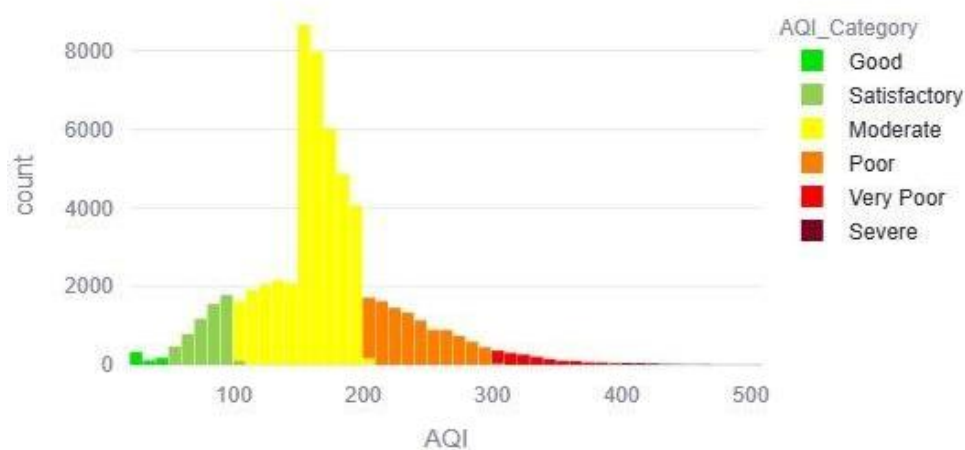


Figure 4.1: AQI Value Distribution by Category (n = 60,291 station-day observations)

4.3 Seasonal Analysis

The seasonal distribution of AQI categories, shown in Figure 4.2, reveals the strong seasonal structure of Delhi's air quality. Winter (December–February) and Post-Monsoon (October–November) seasons are dominated by Moderate, Poor, and Very Poor categories, with a noticeable proportion of Severe episodes during Post-Monsoon. This pattern is driven by the convergence of stubble burning emissions, reduced wind speeds, low planetary boundary layer heights, and temperature inversions that trap pollutants near the surface.

The Monsoon season (July–September) shows the highest proportion of Good and Satisfactory days, reflecting the cleansing effect of rainfall and increased wind speeds.

However, even during Monsoon, the dominant category remains Moderate, confirming that Delhi's air quality is chronically poor even during the most favourable meteorological conditions. Summer (March–June) shows an intermediate pattern, with predominantly Moderate air quality.

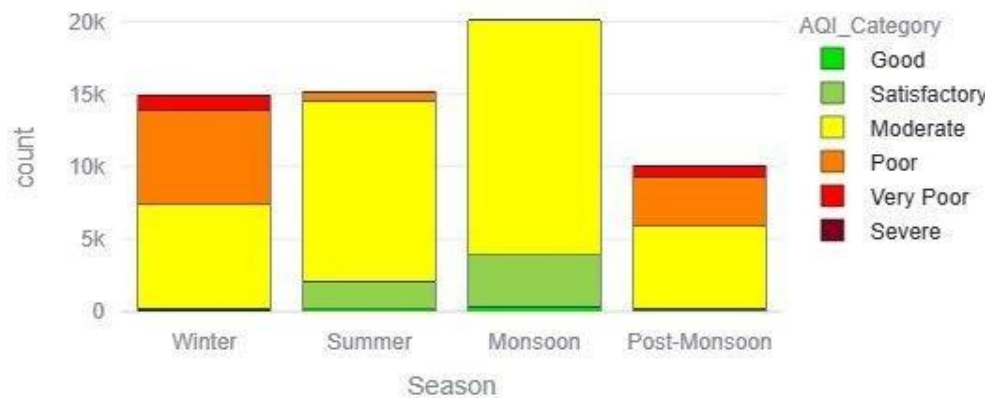


Figure 4.2: AQI Category Distribution by Season (Winter, Summer, Monsoon, Post-Monsoon)

4.4 Yearly Average AQI Trend

Figure 4.3 displays the yearly average AQI from 2020 to 2024. The 2020 average (159) is notably lower than the subsequent years (173 for 2021–2024), attributable primarily to the COVID-19 lockdown of March–May 2020, which dramatically reduced vehicular and industrial emissions across the city. This natural experiment confirms the significant contribution of these anthropogenic sources to Delhi's ambient AQI. The convergence of yearly averages to 173 from 2021 onwards suggests that once pandemic restrictions were lifted, Delhi's air quality rapidly returned to its pre-pandemic baseline, indicating the resilience of the structural pollution sources.

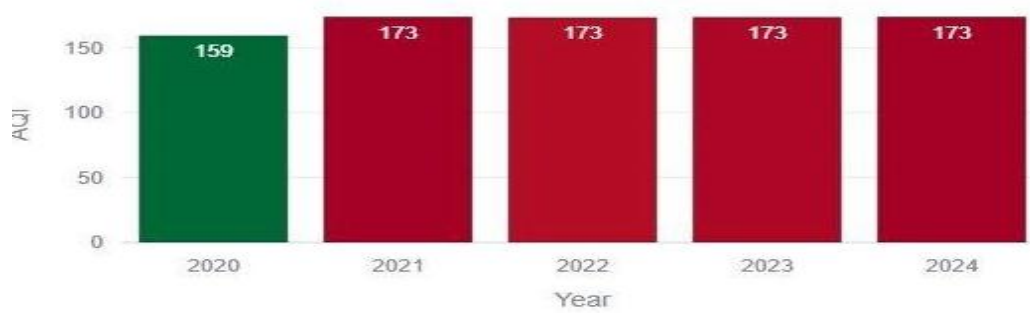


Figure 4.3: Yearly Average AQI (2020–2024); 2020 lower reflects COVID-19 lockdown effect

4.5 Year × Month AQI Heatmap

The Year × Month heatmap in Figure 4.4 provides the most granular view of temporal AQI dynamics across the five-year study period. The heatmap confirms the consistent seasonal pattern: AQI values peak in November–December (orange to deep red, AQI 224–233) and reach their annual minimum in April–June (green, AQI 84–157). The winter peak in November–December is strikingly consistent across all five years, confirming that the seasonal structure of Delhi's pollution is a robust, year-to-year feature rather than an artefact of a single year. The April–2014 corridor showing the lowest AQI (AQI ~84–99) in 2020 stands out clearly, marking the period of the strictest COVID-19 lockdown.

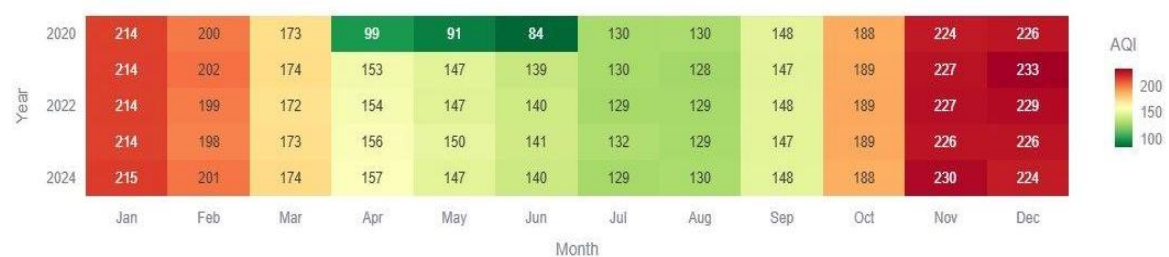


Figure 4.4: Average AQI by Year × Month Heatmap

4.6 Pollutant Correlation Analysis

The Pearson correlation matrix of all six pollutants and AQI, shown in Figure 4.5, reveals the structure of inter-pollutant relationships. PM2.5 and PM10 are very strongly correlated ($r = 0.99$), reflecting their common sources (vehicular emissions, road dust, construction activity) and their co-variation during pollution episodes. Both PM2.5 and PM10 show very high correlations with AQI ($r = 0.97$ and 0.96 , respectively), confirming their dominant role in determining the AQI category. CO also shows a moderately high correlation with AQI ($r = 0.59$), reflecting its role as a tracer of combustion emissions. NO₂ and SO₂ are moderately correlated with AQI ($r = 0.60$ and 0.56 , respectively).

O₃ shows negative correlations with all other pollutants (r ranging from -0.30 to -0.33), reflecting its photochemical production mechanism: ozone is produced from NO₂ and volatile organic compounds in the presence of sunlight, and its concentrations peak during summer afternoons when primary pollutant concentrations are lower. This negative correlation has important implications for machine learning models: the inclusion of O₃ as a feature adds predictive information that is orthogonal to the other pollutants.

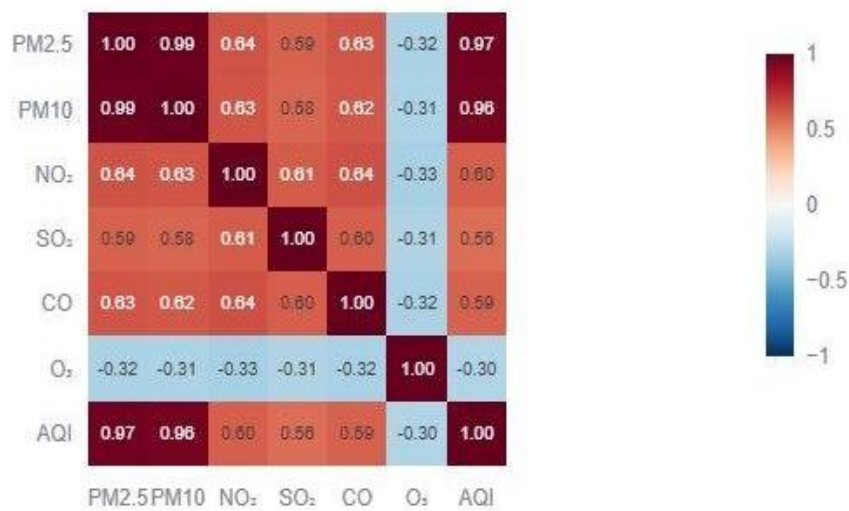


Figure 4.5: Pollutant Correlation Matrix; PM2.5 and PM10 dominate AQI ($r \geq 0.96$)

4.7 Spatial Analysis: Station-Level AQI

Figure 4.6 maps the average AQI at each of the 25 monitoring stations across Delhi. The spatial pattern reveals pronounced heterogeneity: industrial areas and traffic-dense corridors in the northern and western zones of the city (darker red circles) exhibit average AQI values of 182–187, while green belt areas and southern residential zones (darker green circles) have average values as low as 158–162. This spatial gradient of approximately 20–25 AQI units between the most and least polluted stations reflects the localised impact of industrial and traffic emission sources.

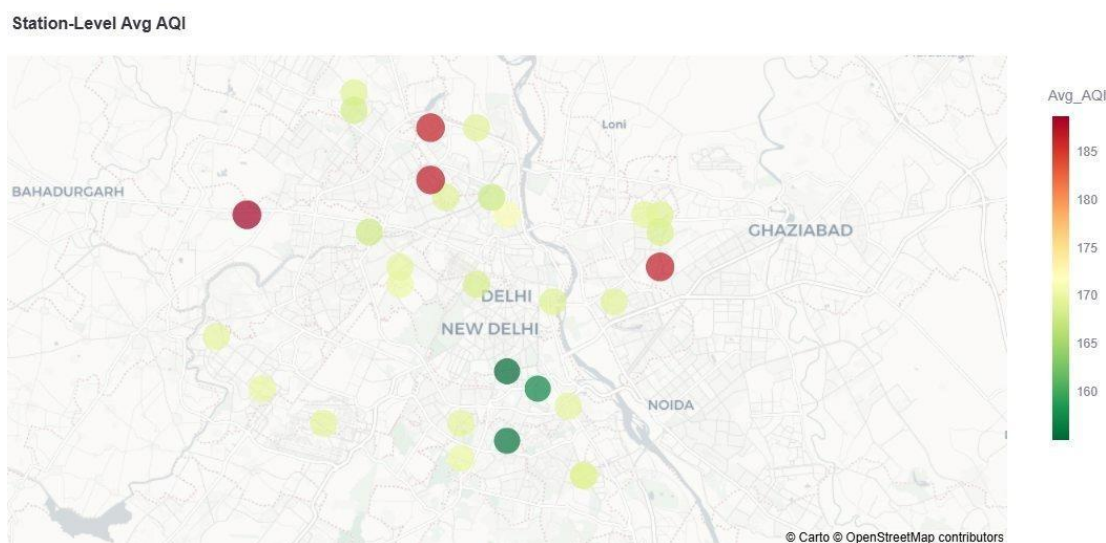


Figure 4.6: Station-Level Average AQI across Delhi

4.8 PM_{2.5}/PM₁₀ Ratio by Station Type

Figure 4.7 shows the PM_{2.5}/PM₁₀ ratio (the "fine particle fraction") by station type. This ratio is a widely used diagnostic indicator of the dominant emission source mix: high ratios (> 0.5) indicate combustion-dominated sources (fine particles from vehicle exhaust, power plants, burning), while lower ratios indicate greater contribution from coarse dust sources (road dust, construction, mechanical abrasion).

Industrial stations show the highest PM_{2.5}/PM₁₀ ratio (0.67), consistent with the dominance of combustion processes in industrial facilities. Airport, Traffic, and Other stations all show ratios of 0.61, above the fine-dominant threshold of 0.50. Green (park) stations show a lower ratio of 0.54, reflecting the greater relative contribution of coarse dust particles in areas away from traffic and industrial sources. All station types exceed the 0.50 fine-dominant

threshold, confirming that combustion is the primary source of particulate pollution at all locations in Delhi.

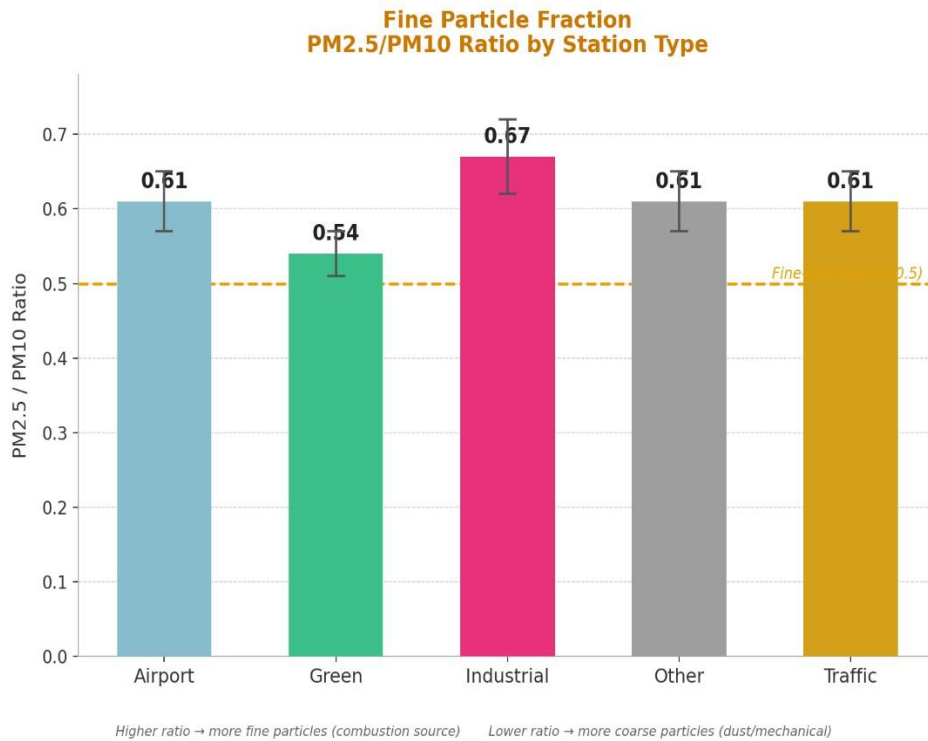


Figure 4.7: PM_{2.5}/PM₁₀ Ratio by Station Type

4.9 Model Performance Comparison

Figure 4.8 and Table 4.2 present the model performance comparison across all five classifiers on the held-out test set of 12,059 observations. Random Forest achieves the highest test accuracy (93.65%, Weighted F1 = 0.9362), followed closely by Logistic Regression (93.49%), Gradient Boosting (93.41%), and Decision Tree (93.40%). These four classifiers are separated by a margin of only 0.25 percentage points, forming a tight performance cluster. KNN (88.48%) is the clear outlier, trailing the cluster by approximately 5 percentage points.

The tight clustering of the top four models is explained by the nature of the AQI classification problem itself. The CPCB AQI is computed as a deterministic, piecewise-linear function of pollutant concentrations, meaning the true decision boundaries between AQI categories are nearly linear in the pollutant feature space. This is why Logistic Regression — despite being the simplest of the five classifiers — achieves 93.49% accuracy. The tree-based ensemble methods (Random Forest, Gradient Boosting, Decision Tree) capture small nonlinearities near category transitions, but their marginal gain over Logistic Regression is only

0.16–0.25%. This confirms that the remaining classification errors stem from genuine boundary ambiguity in the data, not from limitations in model capacity.

KNN’s weaker performance has a clear structural explanation. As a distance-based classifier, KNN assigns equal weight to all features in its Euclidean distance computation. Even after standardisation, this equal-weight metric cannot replicate the implicit feature importance weighting that tree-based models learn during training. In the high-density Moderate AQI region, where observations from adjacent categories are closely interleaved, KNN’s local decision boundaries become unreliable. The result is elevated misclassification in the transitional AQI categories — particularly Satisfactory, Poor, and Very Poor — where distinguishing between classes requires a more structured weighting of pollutant features than KNN can provide.

The mean accuracy across all five models (0.925) confirms that even the weakest classifier correctly identifies more than 9 in 10 observations, underscoring the fundamental learnability of the AQI classification task from concurrent pollutant measurements. The nearperfect alignment of Accuracy, Weighted F1-Score, Precision, and Recall for each model (all within 0.0004 per model) confirms that no classifier exhibits systematic bias toward or against any AQI category on the stratified test set. This consistency across all four evaluation metrics strengthens confidence in the overall robustness of the results.

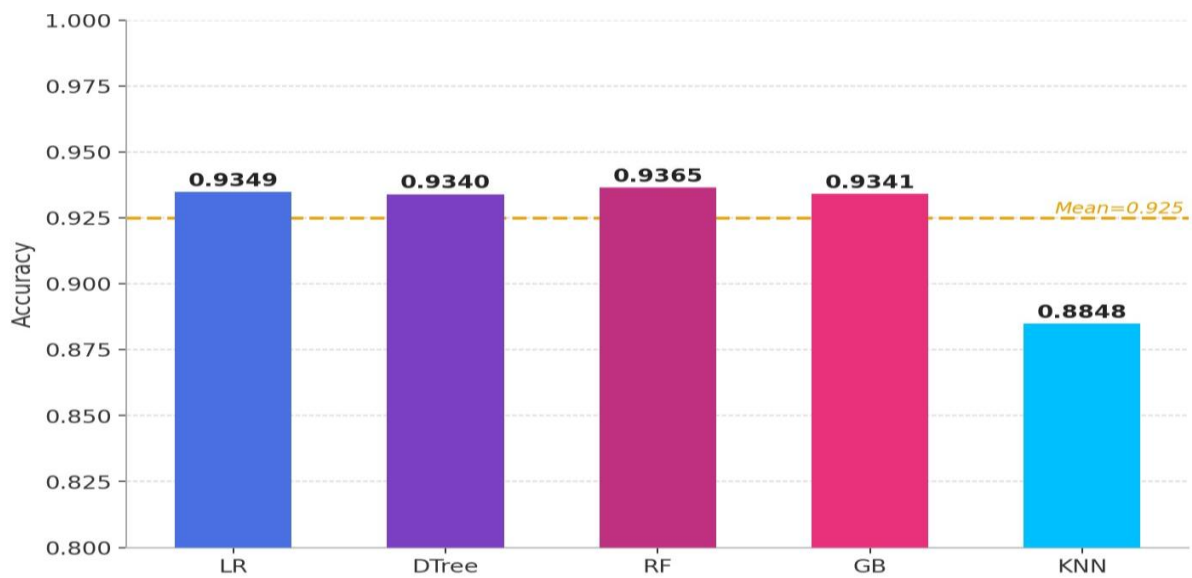


Figure 4.8: Test Accuracy of All Five Classifiers

Table 4.2: Model Performance Summary — Test Set (n = 12,059)

Model	Accuracy	Weighted F1	Precision	Recall
Random Forest	0.9365	0.9362	0.9368	0.9365
Logistic Regression	0.9349	0.9345	0.9352	0.9349
Gradient Boosting	0.9341	0.9338	0.9344	0.9341
Decision Tree	0.9340	0.9336	0.9343	0.9340
KNN (K=5)	0.8848	0.8841	0.8855	0.8848

4.10 Random Forest: Per-Class F1-Scores

Figure 4.9 shows the per-class F1-scores for the Random Forest classifier. Performance is highest for the Moderate category (F1 = 0.965), reflecting its large support in the training data. The Good category achieves F1 = 0.916 and Satisfactory achieves F1 = 0.901, despite their relatively low support. The Severe category shows the lowest F1-score (0.652), reflecting both its low support (approximately 2% of observations) and the greater ambiguity of boundary observations between the Very Poor and Severe categories. The Very Poor category achieves F1 = 0.720. These results highlight that the primary remaining challenge for AQI classification in Delhi is the reliable identification of the most severe pollution episodes.

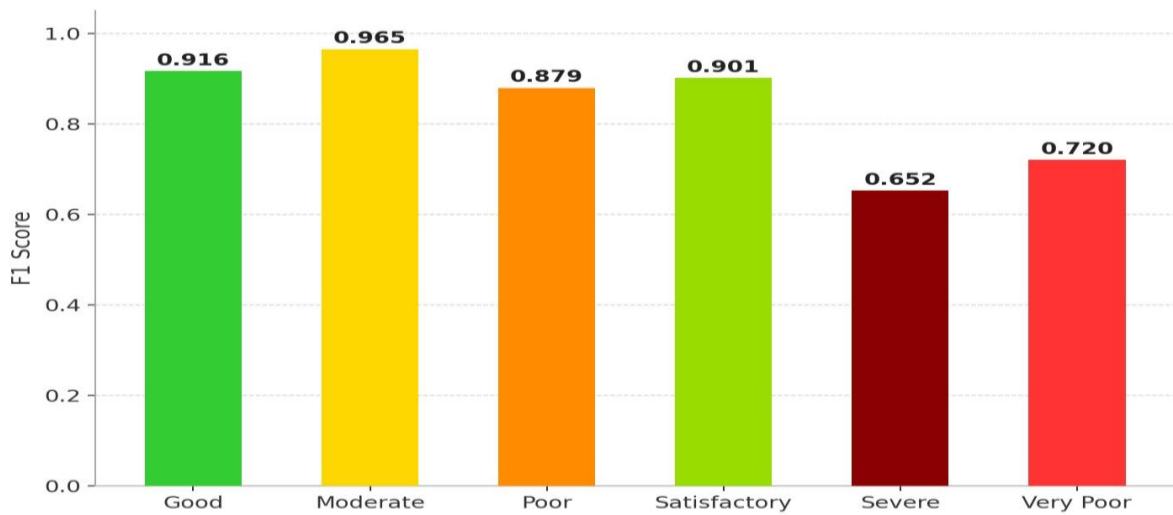


Figure 4.9: Random Forest Per-Class F1-Scores; Lowest performance on Severe (0.652) reflects class imbalance

4.11 Feature Importance Analysis

Figure 4.10 presents the Mean Decrease in Impurity (MDI) feature importance estimates from the Random Forest classifier. PM10 emerges as the dominant predictor with an importance score of 0.7505, reflecting its very high correlation with AQI ($r = 0.96$) and its role as the primary sub-index pollutant on a majority of days in Delhi. NO₂ is the second most important feature (0.0762), followed by CO (0.0554), SO₂ (0.0488), Month (0.0293), O₃ (0.0257), and Year (0.0142).

The high importance of Month (rank 5) confirms the strong seasonal structure of AQI categories documented in the EDA, and validates the inclusion of the calendar month as a temporal predictor. The non-trivial importance of O₃ (0.0257) is consistent with its negative correlation with other pollutants: its presence provides information about the photochemical regime, which is complementary to the primary pollutant concentrations. The Year feature has the lowest importance (0.0142), suggesting that there is no strong year-to-year trend in AQI category composition beyond what is captured by the seasonal and pollutant concentration features.

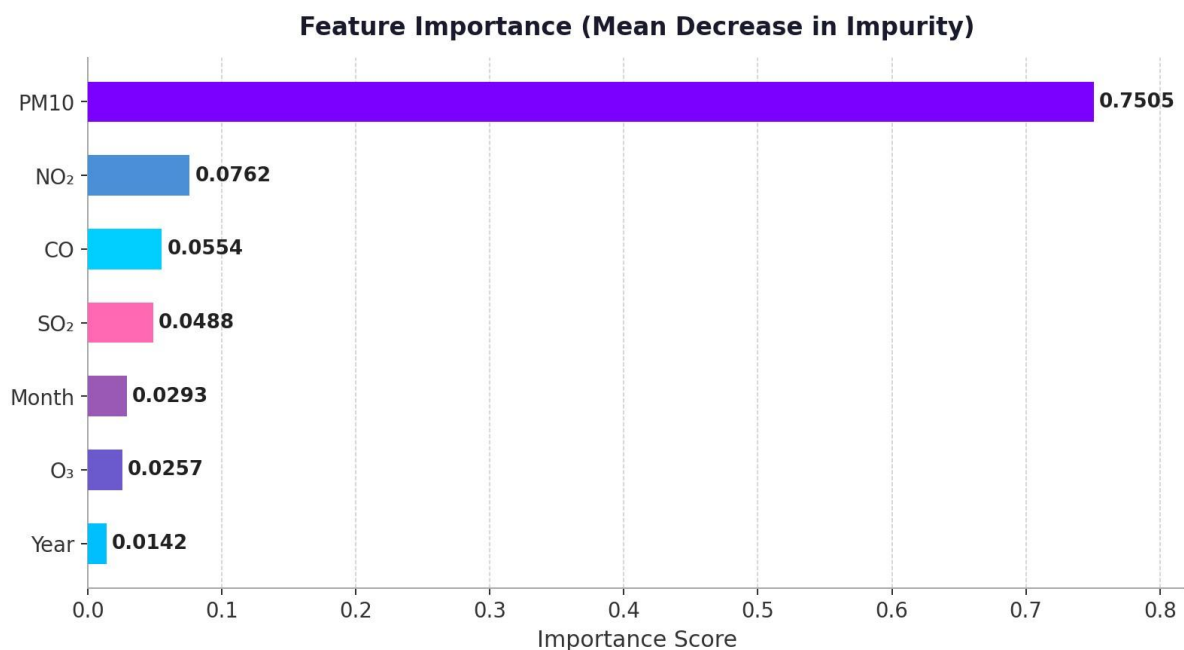


Figure 4.10: Feature Importance (Mean Decrease in Impurity) — PM10 dominates with importance 0.7505

CHAPTER V

SUMMARY AND CONCLUSION

5.1 Summary of Findings

This study presented a comprehensive machine learning-based framework for the automated classification of Air Quality Index categories in Delhi, India, using a large-scale multi-station dataset of 60,291 station-day observations spanning January 2020 to December 2024. The study evaluated five supervised classifiers — Random Forest, Gradient Boosting, Logistic Regression, Decision Tree, and K-Nearest Neighbours — on the task of predicting six AQI categories from concurrent measurements of six pollutants (PM_{2.5}, PM₁₀, NO₂, SO₂, CO, O₃) and the calendar month.

The principal findings are summarised as follows:

- Delhi's air quality exhibits strong, consistent seasonal patterns. AQI values peak in November–February (winter), driven by stubble burning, reduced wind speeds, and temperature inversions. Monsoon (July–September) provides a natural cleansing period. These patterns are highly consistent across all five years.
- The Moderate AQI category (101–200) is the dominant category (~49.7% of observations). The Good category (<50) accounts for only ~1%, confirming chronic poor air quality even during relatively clean periods.
- PM₁₀ is the dominant predictor of AQI category (Random Forest importance = 0.75), consistent with Delhi's profile as a particulate-dominated pollution environment. The PM_{2.5}/PM₁₀ ratio exceeds 0.5 at all station types, confirming combustion as the primary fine particle source.
- Random Forest achieves the highest test accuracy (0.9365), followed by Logistic Regression (0.9349), Gradient Boosting (0.9341), Decision Tree (0.9340), and KNN (0.8848). All ensemble and linear models exceed 93% accuracy.
- Per-class F1-scores from Random Forest are highest for Moderate (0.965) and lowest for Severe (0.652), reflecting the challenge of classifying rare extreme pollution episodes.

- The COVID-19 lockdown of March–May 2020 produced a measurable reduction in Delhi's average AQI (159 vs. 173 in subsequent years), providing a natural experiment that quantifies the contribution of vehicular and industrial emissions.

5.2 Conclusions

The results of this study lead to the following conclusions. First, machine learning classifiers — particularly Random Forest — can predict AQI categories from concurrent pollutant measurements with accuracy exceeding 93%, making them suitable for real-world deployment in automated air quality monitoring systems. The strong performance is attributable to the high correlation between particulate matter concentrations and the CPCB AQI sub-index computation, which effectively maps pollutant concentrations to AQI categories through a deterministic rule that ML models can closely approximate.

Second, the competitive performance of Logistic Regression (93.49%) alongside ensemble methods confirms that the fundamental AQI category decision boundaries are approximately linear in the pollutant concentration space. This has practical implications for deployment: in environments where computational resources are constrained or model interpretability is paramount, Logistic Regression provides a viable alternative to Random Forest with only marginal accuracy loss.

Third, the temporal analysis demonstrates that Delhi's air quality has not shown a significant improving trend over 2020–2024 at the city-wide scale. The sharp AQI reduction during the COVID-19 lockdown, followed by rapid return to baseline, confirms that the structural drivers of pollution — vehicular emissions, industrial activity, biomass burning — are persistent and will reassert themselves unless targeted, sustained policy interventions are implemented.

Fourth, PM₁₀ dominates the feature importance analysis (0.7505), confirming that coarse and fine particulate matter is the primary driver of AQI category classification in Delhi. Policy interventions targeting particulate matter sources — including vehicle emission standards, industrial dust control, and construction site regulations — are therefore of the highest priority for air quality improvement.

BIBLIOGRAPHY

- 1.Arsov, M., Zdravevski, E., Lameski, P., Corizzo, R., Koteli, N., Mitreski, K., & Trajkovik, V. (2020). Short-term air pollution forecasting based on environmental factors and deep learning models. In *2020 15th Conference on Computer Science and Information Systems (FedCSIS)* (pp. 15–22). IEEE.
- 2.Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
- 3.Central Pollution Control Board (CPCB). (2014). *National Air Quality Index*. Ministry of Environment, Forest and Climate Change, Government of India.
- 4.Chaudhuri, S., & Dutta, D. (2014). Mann–Kendall trend of pollutants, temperature and humidity over an urban station of India with forecast verification using different ARIMA models. *Environmental Monitoring and Assessment*, 186(8), 4719–4742.
- 5.Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5), 1189–1232.
- 6.Gopu, P., Panda, R. R., & Nagwani, N. K. (2021). Time series analysis using ARIMA model for air quality prediction in Hyderabad city of India. In *Soft Computing and Signal Processing: Proceedings of 3rd ICSCS 2020* (pp. 47–56). Springer.
- 7.Liu, T., & You, S. (2022). Analysis and forecast of Beijing's Air Quality Index based on ARIMA model and neural network model. *Atmosphere*, 13(4), 512.
- 8.McKinney, W. (2010). Data structures for statistical computing in Python. In *Proceedings of the 9th Python in Science Conference*.
- 9.Pakrooh, P., & Pishbahar, E. (2019). Forecasting air pollution concentrations in Iran using a hybrid model. *Pollution*, 5(4), 739–747.
- 10.Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., & Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

- 11.Samia, A., Kaouther, N., & Abdelwahed, T. (2012). A hybrid ARIMA and artificial neural networks model to forecast air quality in urban areas: Case of Tunisia. In *Advanced Materials Research* (Vol. 518, pp. 2969–2979). Trans Tech Publications.
- 12.Siew, L. Y., Chin, L. Y., & Wee, P. M. J. (2008). ARIMA and integrated ARFIMA models for forecasting air pollution index in Shah Alam, Selangor. *Malaysian Journal of Analytical Sciences*, 12(1), 257–263.
- 13.Slini, T., Karatzas, K., & Moussiopoulos, N. (2002). Statistical analysis of environmental data as the basis of forecasting: An air quality application. *Science of the Total Environment*, 288(3), 227–237.
- 14.World Health Organization. (2021). *WHO global air quality guidelines: Particulate matter (PM_{2.5} and PM₁₀), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide*. WHO.
- 15.E3S Web of Conferences. (2019). *E3S Web of Conferences* (Vol. 136, Article 05001). EDP Sciences.